EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.1 Fact-Finding Techniques

Following project initiation, to gain a deep understanding of the specific pain points and potential requirements of the existing Teaching Assistant (TA) recruitment process at BUPT International School, we selected **User Interviews** as the primary and most direct requirements elicitation method. We conducted semi-structured, in-depth interviews with the two most critical stakeholder groups of the system: **Teaching Assistants (TAs)** and **Module Organisers (MOs)**. This approach allowed us to explore open-ended questions, capturing users' emotions, motivations, and unstated needs, which is crucial for understanding a new system designed to replace the existing manual process.

1.1.1 Interview with Teaching Assistants (TAs):

We interviewed a student with prior TA application experience. The interview revealed multiple inefficiencies and information barriers in the current process. Key findings include:

1) Information Opacity:

When applying, TAs can only see a list of course names. They cannot access specific details such as the required skills for the course, the name of the Module Organiser (MO), or the number of applicants. This leads to application decisions being made somewhat blindly.

2) Fragmented and Cumbersome Process:

The application process involves several disjointed steps. First, TAs fill out a form with personal information via Microsoft Forms. Subsequently, they must **manually** rename their CV files to a specific format and send them separately via email to a designated coordinator. This process is prone to error and adds cognitive load for applicants.

3) Lack of Feedback Mechanism:

After submitting an application, applicants are in an "information vacuum." They cannot know if their application has been reviewed, if they have been waitlisted, or rejected. The only notification is receiving a direct offer email from the teacher; rejected applicants typically receive no notification at all.

4) Inefficient Communication:

Even upon receiving an offer, accepting or rejecting it still requires a reply via **email**, failing to close the loop within a unified platform.

5) No Systematic Workload Tracking:

Once hired, a TA's workload (e.g., working hours) lacks systematic recording. Currently, reporting is done informally, such as through WeChat group messages, which is not conducive to management and statistics.

1.1.2 Interview with Module Organisers (MOs):

We interviewed a course instructor who recruits TAs, focusing on the workload of the hiring side. The interview clarified that the MO's core need is to simplify the recruitment process and improve screening efficiency. Key findings include:

1) High Burden of Information Filtering: Currently, an MO receives a large Excel spreadsheet containing all applicants. The teacher must manually sift through this to find students who applied for their specific course, an unnecessary and time-consuming step. The MO explicitly stated, "I actually only want to see people who applied for my course. I don't want to look at information for hundreds of applicants."

2) Unfriendly Interface Display:

MOs prefer to view applicants in a clear **list format**, allowing quick scanning of key information, rather than a "one person per page" format requiring page flipping, which significantly reduces screening efficiency.

3) Cumbersome Offer and Communication Process:

After selecting candidates, the MO must **manually** find the student's email address from the CV or spreadsheet and then send individual offer emails. If a candidate declines, this process must be repeated to find the next candidate. The teacher described this as "particularly troublesome."

4) Clear Process Boundaries:

MOs prefer to review all applications uniformly after the application deadline and want the ability to manually close the application channel once positions are filled, to avoid subsequent interruptions. Furthermore, MOs explicitly stated they would not consider students who did not apply for their course, and once an applicant is rejected, that applicant should not reappear in subsequent candidate lists.

Furthermore, to gain a more complete understanding of the existing process, we also

analyzed the official Teaching Assistant application form currently in use (a questionnaire distributed via Microsoft Forms). The structure and fields of this form (such as the maximum number of course selections, personal information collection items, etc.) directly corroborate some procedural details mentioned by users during the interviews (e.g., "can only choose three courses"). It also provided concrete reference points for designing the data model and user interface of our new system. A screenshot of this form is included in the supporting materials appended to this report.

1.2 Iteration Plan

Adhering to Agile development principles and following the suggested timeframe in the course handout, we have planned the project development into four iterative Sprints. Our **Product Backlog** has already defined the goals and user stories to be completed in each iteration in detail, ensuring that each sprint delivers a working, valuable software increment.

Sprint 1: Core User Management & System Foundation (BUPT Week 3-4).

The goal of this sprint is to establish the foundational framework and core authentication mechanism of the system. We will implement account registration and login for all three user types (TA, MO, Admin) and enforce strict **Role-Based Access Control (RBAC) (SYS-001)**, ensuring users can only access features within their permissions. Concurrently, we will implement the basic TA profile creation functionality to prepare data for the subsequent application process. This forms the cornerstone of the entire system.

Sprint 2: Core Recruitment Workflow Implementation (BUPT Weeks 5-6).

This sprint will deliver the core value of the system, addressing the main pain points identified in the user interviews. Features include: MO posting jobs with detailed requirements (MO-003); TA browsing available jobs, viewing job details (TA-004, TA-005), and applying (max 3 applications, TA-006); TA uploading CVs (TA-007); MO viewing a list of applicants for their own courses (MO-004) and reviewing them (Accept/Reject/Waitlist, MO-005); MO sending offers with one click via the system (MO-006). Simultaneously, TAs can check their application status in real-time (TA-008) and directly accept or reject offers within the system (TA-009). Upon completion of this sprint, a usable end-to-end recruitment workflow will be essentially formed.

Sprint 3: Feature Expansion & Process Optimization (BUPT Weeks 7-8).

With the core workflow stable, this sprint will expand the system's management and auxiliary features to enhance user experience and system completeness. This includes: TA viewing personal workload records (TA-010); MO managing contact information of hired TAs (MO-008) and manually closing recruitment positions (MO-009); Admin viewing global data, managing application periods, and user accounts (ADM-002, ADM-003, ADM-004); and the process for MOs to quickly select replacements from the waitlist if a TA declines an offer (MO-007).

Sprint 4: Integration, Testing & Delivery Preparation (BUPT Weeks 9-10).

This is the final sprint, shifting focus from feature development to product refinement. We will conduct full system integration testing, fix defects, and refine the user interface and performance. Concurrently, we will strictly adhere to the course handout requirements to prepare the final deliverables: the complete, runnable software, demonstration video, user manual, test report, and final project report.

1.3 Prioritisation Methods

To scientifically manage the order of feature development, we employed the **MoSCoW prioritisation method** to categorize all user stories. The primary basis for prioritization decisions was the **frequency and severity of pain points revealed in user interviews**, as well as the set of features essential to building a **Minimum Viable Product (MVP)**. All priorities are recorded in the "Priority" column of the Product Backlog.

1) Must Have:

These are the foundational elements of the system. The absence of any would prevent the system from supporting the most basic recruitment workflow. They directly target the most critical and time-consuming pain points in the current manual process. For example, TA applying for jobs and MO posting and reviewing jobs are the starting point and core decision points of the process; Role-based access control is a prerequisite for system security and data segregation; Sending and replying to offers within the system directly eliminates the cumbersome back-and-forth emails. Together, these features constitute the MVP that solves the users' core problems.

2) Should Have:

These features build upon the MVP, significantly enhancing the system's manageability, completeness, and user experience, but the core business process can still function without them. Examples include administrator functions, workload tracking, and more refined recruitment process management (e.g., closing positions, handling waitlists). These will be implemented in later iterations to enrich the product's value.

1.4 Estimation Methods

To forecast the workload achievable per sprint and plan effectively, we adopted the **Story Points Estimation** method. The team, through planning poker sessions, assessed the relative size, complexity, and uncertainty of each user story. We used a modified Fibonacci sequence (1, 2, 3, 5, 8) as the unit of points, with the results recorded in the "Estimation" column of the Product Backlog.

Estimation considers not only coding time but also the effort for requirements analysis, design, testing, and integration. For example, "TA Profile Creation" (TA-003) and "MO Post

Job Openings" (MO-003) were estimated as 5 points due to their higher complexity involving multiple data fields, input validation, and file operations. Simpler foundational features like "TA Login" (TA-002) were estimated as 3 points. By combining user story points with the team's historical velocity (to be established after Sprint 1), we can more reliably plan the deliverable content for each iteration, ensuring the commitments are achievable.

2. Supporting material

2.1 Transcript of TA Interview (13 March 2026)

Interview Time: *March 13, 2026*

Interviewee: *Teaching Assistant(TA)*

Interviewer: *Student members of the project team*

Interview Transcript:

Interviewer: How is the TA application process currently conducted?

TA: Right now, a QR code or link is posted on Teams. Clicking it leads to a Microsoft Forms survey, which is Microsoft's data collection form. TAs need to fill out this form, which collects all the application information and automatically consolidates it into an Excel spreadsheet. After that, the instructor will look at all the applicants' information in this Excel spreadsheet and screen for suitable candidates.

Interviewer: Can TAs choose courses when filling out the form?

TA: You can choose courses, but with a maximum of three courses. However, when filling it out, you don't actually know which course you'll end up being accepted into, and you don't know the competition. There's no feedback.

Interviewer: If the instructor gives you an offer, you should receive a notification, right?

TA: Strictly speaking, yes, I can receive notification. But that happens via email. If I want the course, I reply to their email; if I don't want it, I should also reply to their email. Yes, that's the process.

Interviewer: What kind of feedback would you hope to get during this process? For example, if the instructor is interested in you, adds you to a shortlist, and you could see that you've been waitlisted—would that count as feedback for you?

TA: That would definitely be more information. Right now, there's no signal. It's just a direct email, and they say, "You must tell us if you can't come."

Interviewer: If you're rejected, would you like a chance to apply for other courses?

TA: Of course. Right now, I can only choose 3. If I get rejected from all of them, I can't be a TA anymore. If I could choose other instructors, I'd definitely want to.

Interviewer: You said the instructor checks all applicants' information in the Excel spreadsheet and screens for suitable candidates.

TA: Right. If an instructor needs a TA, they have to look at a massive Excel spreadsheet. And if you select someone, and that person has also selected another course causing a conflict, that gets very messy. The instructor might have to look at the table again.

Interviewer: When the instructor picks someone, do they only pick from people who chose their course, or from everyone?

TA: They can see everyone's information, and then they ask you if you're willing. Currently, it's—they post the job opening, you just fill in the form and submit it, and then you wait for an email. If multiple instructors email you, you reply to them—I accept, or I don't accept.

Interviewer: So currently, you just fill out the form directly? Do you need to provide a CV? What information is needed for the CV?

TA: The CV—it's like this: We rename our CV in a specific way, we have to manually change the name, for example, to StudentID_Name, and then send it to someone's email. That person compiles all the CVs together, and then the instructor might see those CVs.

Interviewer: So, you fill out the form first, and then the CV is sent separately?

TA: Yes, the CV is sent separately to one person. That person might do some work later.

Interviewer: Would it be better if uploading the CV was required when filling out the form?

TA: Mm-hmm.

Interviewer: Do you know roughly what the job entails? What skills you need to have?

TA: No.

Interviewer: You don't know the specific requirements either?

TA: I don't even know who teaches the course because I'm not in that major.

Interviewer: So you can only see the course name?

TA: Right. I roughly think—I seem like I can do it, so I choose it.

Interviewer: But when they write the requirements, don't they clearly state how many people are needed, what skills are required?

TA: When I fill out the form… I don't really remember. It's just A, B, C, D, such and such a course, a name after the course, and you just choose.

Interviewer: For example, if the course requirements are given to you in advance, and you tailor your CV based on those requirements, wouldn't that increase your chances of being selected?

TA: I'd definitely prefer that. Think about it, if I specifically write things unrelated to the course? If I leave out some of my experience, then I probably won't get it, right?

Interviewer: Now that you are a TA, is your workload—the hours you work—recorded? Is there a clock-in system?

TA: I might need to… I need to send some messages in the WeChat group. For example, I'll post all the questions you asked me today in the group, and then… that counts as me being present, right?

Interviewer: This indicates that you have completed your work?

TA: Yes.

2.2 Transcript of MO Interview (13 March 2026)

Interview Topic: *Interview on Pain Points and Suggestions for Improving the Teaching Assistant Application System*

Interview Time: *March 13, 2026*

Interview Location: *Offline discussion*

Interviewer: *Project team members*

Interviewee: *Module Organiser (MO)*

Interview Transcript:

Interviewer: When you post a TA recruitment notice, do you have specific requirements, such as needed skills?

MO: Yes, of course. What skills are needed, working hours, when it starts, when it ends. If you meet the criteria, you apply. Of course, anyone can apply.

Interviewer: In the current TA application system, how do instructors view application information?

MO: The current system allows you to see all applicants' information. The instructor has to screen it themselves, see which students applied for their course, and then look at their resumes one by one.

Interviewer: So, the MO needs to screen from all the applicants themselves?

MO: Yes. Actually, I only want to see the people who applied for my course. I don't want to look at information for hundreds of applicants. If I only need 15 TAs, and maybe about 20 people applied, then I just need to pick 15 from those 20.

Interviewer: If most MOs are only concerned with the students who chose their course, then our system should simply let you see only those people, without needing to screen yourself?

MO: Right, no need to screen myself. I don't want to look at other people's courses. What do other people's courses have to do with me?

Interviewer: What if the positions aren't filled?

MO: We'll deal with that if it happens. The first step, I just want to see the applicants for my course.

Interviewer: How do you notify them after you find suitable people now?

MO: It's very troublesome now. After I find the resume, I have to look for their email, and then email them to notify them one by one.

Interviewer: What if someone refuses?

MO: Find someone else, send another email. It's all manual work.

Interviewer: Would you select from people who didn't apply for your course?

MO: I don't want to select from people who didn't apply for my course. If they haven't applied, it means they're not interested. Why would I select from there? The MO is always the active party. Students who have applied for a course can become candidates. If you didn't apply, I can't transfer you. Because the skill requirements for each course are very specific—this course involves microprocessors, you must know assembly programming; that course might require knowledge of law. Completely different.

Interviewer: What if we implement keyword matching? For example, the system automatically recommends candidates with higher keyword matching between the skills you list and the student's CV?

MO: It's possible... but I don't need it. Because this isn't large-scale recruitment—it's not like I have thousands of applications needing AI to filter. I need 15, and usually only about 20 people apply. It's actually easier for me to look at them myself. I don't need the system to handle this. Making it too complicated isn't good. How can I trust the matching is reasonable? Maybe the matching misses a better candidate. For a large company with thousands or tens of thousands of applications, you have to use filters. But with our scale, it's really only about 20, not more.

Interviewer: Regarding the way information is displayed, is it better to put everyone's information on one page and scroll, or to have one person per page to flip through?

MO: I just want a list. I don't want one person per page; 20 people, flip 20 pages? I don't want that. I just want a list, clear and simple, these 20 people.

Interviewer: Is there a deadline? TAs submit before that, and then you review them all at once?

MO: Yes, there's a deadline. After the deadline, I review them all at once, and once I have enough, I close it.

Interviewer: If the first round isn't full, and a second round is done, can you see people you rejected in the first round again?

MO: If rejected, I definitely don't want them. Rejecting them means they don't meet my requirements. I'd rather be short-staffed than take them. So, I shouldn't see them again.

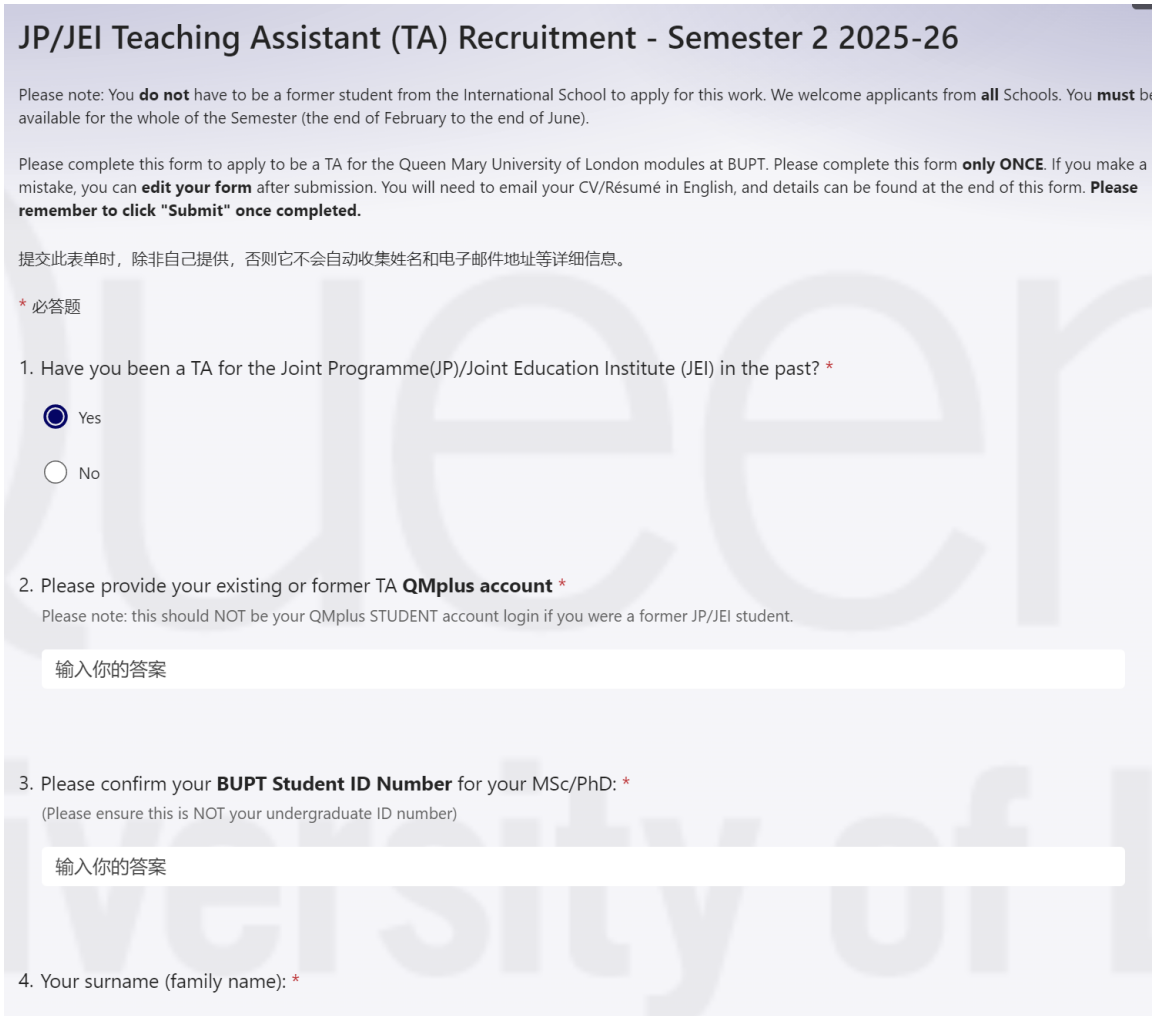
Interviewer: The system should automatically filter them out?

MO: Right, they shouldn't appear anymore. But to be honest, this situation is rare. You should focus on the most basic functions first—people like me needing 15 TAs are a very small minority. Most courses only need one or two TAs. Don't design overly complex things for a special case like mine.

Interviewer: Okay, thank you!

MO: You're welcome.

2.3 Screenshot of the Current TA Application Form (Microsoft Forms, Semester 2 2025–26)



JP/JEI Teaching Assistant (TA) Recruitment - Semester 2 2025-26

Please note: You **do not** have to be a former student from the International School to apply for this work. We welcome applicants from **all** Schools. You **must** be available for the whole of the Semester (the end of February to the end of June).

Please complete this form to apply to be a TA for the Queen Mary University of London modules at BUPT. Please complete this form **only ONCE**. If you make a mistake, you can **edit your form** after submission. You will need to email your CV/Résumé in English, and details can be found at the end of this form. **Please remember to click "Submit" once completed.**

提交此表单时，除非自己提供，否则它不会自动收集姓名和电子邮件地址等详细信息。

* 必答题

1. Have you been a TA for the Joint Programme(JP)/Joint Education Institute (JEI) in the past? *
   - (●) Yes
   - ( ) No

2. Please provide your existing or former TA **QMplus account** *
   Please note: this should NOT be your QMplus STUDENT account login if you were a former JP/JEI student.
   输入你的答案

3. Please confirm your **BUPT Student ID Number** for your MSc/PhD: *
   (Please ensure this is NOT your undergraduate ID number)
   输入你的答案

4. Your surname (family name): *

4. Your surname (family name): *

输入你的答案

5. Your forename (given name): *

输入你的答案

6. Your Chinese name: *

输入你的答案

7. Your mobile/cell phone number: *

输入你的答案

8. What is your gender? *

- ◯ Female
- ◯ Male

9. Which School are you from at BUPT? *

输入你的答案

10. Please provide the **name** of your Supervisor at BUPT *

Enter their full name

输入你的答案

11. Are you **currently** a Masters (MSc) or a PhD student? *

- ◯ Masters (MSc)
- ◯ PhD

12. Which **Year** of your course are you **currently** in? *

- ◯ Year 1
- ◯ Year 2
- ◯ Year 3
- ◯ Year 4
- ◯ Year 5

13. Have you ever been a student on the Queen Mary-BUPT Joint Programme (JP)/Joint Education Institute (JEI)? *

If "Yes", what was your programme/major (e.g. Internet of Things Engineering) and what year did you graduate?

输入你的答案

14. Which campus are you based at? *

- [ ] BUPT Main (XiTuCheng) Campus
- [ ] BUPT Shahe Campus

15. Please select any of the following skills or areas of knowledge you have (you can only select a maximum of 3 modules, so please choose carefully): *

请最多选择 3 个选项。

- [ ] (EBU4202) Knowledge of combinational and sequential logic devices and circuits and the associated laboratory test equipment (very important).
- [ ] (EBU4211) Knowledge of Java (basic concepts of fundamentals of Java, Object-oriented Java, Java libraries, use of Java IDEs and coding in Java) and Knowledge of Python (Object Oriented Programming using Python, basic concepts of Python and their differences with Java concepts, Python functions and libraries for data handling).
- [ ] (EBU5315) Experience in HTML, CSS, JavaScript.
- [ ] (EBU5376) Familiar with Fourier Transform, frequency spectrum, and digital filter design. Have knowledge of Matlab.
- [ ] (EBU5408) Audio processing and digital signal processing - covering audio effects, signal quantisation, FFTs, and filtering - along with MATLAB and Python skills.
- [ ] (EBU5477) C programming; preferably some knowledge on MCS51 or ARM microcontrollers, peripherals - UART, I2C, PWM - and interrupt; digital signals and systems, digital logic design, analogue circuits and design; ADC/DAC/Filters.
- [ ] (EBU5503) Relational database, ER model - be able to create ER diagram from given requirements - Relational model - able to map ER diagram to a relational model - SQL, MySQL - Workbench and command line.
- [ ] (EBU5606) Knowledge of product design and development, including fundamental knowledge of marketing principles.
- [ ] (EBU6230) Fundamental knowledge of image processing - how image manipulation and enhancement works - and the associated techniques e.g., filters and must have good Matlab and Python skills.
- [ ] (EBU6304) Java Programming; Software Development; Agile Process.

- [ ] (EBU6304) Java Programming; Software Development; Agile Process.
- [ ] (EBU6305) Experience in HTML, CSS, JavaScript
- [ ] (EBU6307) Experience in python programming, GPU access and usage and use of packages like scipy, sklearn, numpy, pandas, cv2, and deep learning packages such as tensorflow, pytorch required. Must also know about fundamentals of signals/image processing, computer vision and machine learning.
- [ ] (EBU6366) Knowledge of microwave engineering. Experience in CST simulation, and experiments with vector network analyzer.
- [ ] (EBU6408) Knowledge of communication and signal processing theories, and preferably knowledge of RFID systems. Familiarity with C programming, preferably knowledge/experience with Arduino development kits.
- [ ] (EBU6475) Knowledge of ARM architecture, mixed C and Assembly programming, STM32 - ARM - HAL programming, peripherals - UART, I2C, PWM - and interrupt.
- [ ] (EBU6505) Understanding of AI Concepts in reasoning, search algorithms, reinforcement learning, and probabilistic reasoning. Experience with Python and tools like Numpy, pandas, scikit-learn, PyTorch, or libraries for knowledge representation, Bayesian networks, or reinforcement learning. Experience with LLM and agent development framework, e.g., LangChain, LangGraph, etc.
- [ ] I am available to help at the Shahe Campus to support extra-curricular activities, such as talks and events.

16. You now need to **email** ONE version of your CV/Résumé in English as a PDF or Word file attachment. The file name of your CV/Résumé must be in the format "BUPT ID_SURNAME_Forename (in pinyin)", for example: **2021123456_LIU_Xiao**

Please email your CV/Résumé **as an attachment** to the email address **qmbupt@qmul.ac.uk** only once.

Once you have emailed your CV/Résumé in English with the correct filename to **qmbupt@qmul.ac.uk**, click the option below and then click "Submit" to send this application form. *

- ( ) I have emailed my CV with the correct filename

你可以在提交后打印你的答案的副本

提交